

Conclusion

Interstitial CT-guided BT is an effective treatment option with a good rate of local control. However, longer follow-up is needed to definitively assess its role in this setting.

OC-0114 Results from the First Spanish National Survey on Epiescleral Brachytherapy for Uveal Melanoma

I. Rodriguez Rodriguez^{1,1}

¹*La Paz Hospital, Radiation Oncology, Madrid, Spain*

Purpose or Objective

Perform a national survey to establish the use of episcleral brachytherapy as a treatment option for uveal melanoma over a 10-year period and describe patient and tumour characteristics, work-up studies, dosimetry and planning, as well as patient follow-up and treatment complications.

Materials and Methods

Seven reference hospitals at the national level with considerable experience in episcleral brachytherapy for uveal melanoma (performance of at least 10 implants per year) participated in the study. The survey was carried out using Google Forms with 42 questions, and the resulting descriptive statistics were done in Microsoft Excel. Episcleral brachytherapy was performed with *ROPES*, Collaborative Ocular Melanoma Study (*COMS*) and *Eckert & Ziegler BEBIG's* plaques loaded with Iodine¹²⁵ seed models 6711 y 125.S16, and Rutenio¹⁰⁶. Planning followed American Association of Physicists in Medicine (AAPM), American Brachytherapy Society (ABS) and COMS guidelines.

Results

Between 2007 and 2017, 1350 patients were treated with episcleral brachytherapy, a mean of 25 patients' year. All centres used echography and ophthalmoscopy for diagnosis. According to the COMS classification, 17%, 70% and 13% reported tumours were small, medium and large respectively. Median age was 65 years old; 54,2% were women and 45,8% were men. Five out of seven centres used both I¹²⁵ and Ru¹⁰⁶. Dosimetry and three-dimensional reconstruction were performed by a computer system developed by Dr. Astrahan at the University of California, using version 2.16. The clinical target volume was defined through echography with a 2mm margin around tumor base. Prescription dose was to the tumour apex. Dosimetry for I¹²⁵ was calculated using COMS. Dose prescription was 85Gy for both I¹²⁵ and Ru¹⁰⁶. Maximum and minimum dose for I¹²⁵ treatment ranged from 0,4Gy/hour to 1,25Gy/hour, which implied five to seven-day hospitalization. All centres performed patient follow-up.

Conclusion

This was the first Spanish national survey performed to determine the use of episcleral brachytherapy for uveal melanoma. Although the treatment is highly defined, there are variations among centres on which it would be convenient to reach a consensus. A closer patient follow-up is highly needed to achieve accurate results in terms of local control rate and brachytherapy side-effects.

OC-0115 Detector-based quality assurance of assembled brachytherapy eye plaques

S. Moehle¹, I. Spadinger¹

¹*BC Cancer - Vancouver Center, Department of Medical Physics, Vancouver, Canada*

Purpose or Objective

This study presents an experimental dosimetry system for pre-insertion quality assurance of assembled brachytherapy eye plaques using a high-resolution CMOS flat-panel x-ray detector. A calibration method using Monte Carlo (MC) simulations is established and experimental results are compared to MC calculations for various plaque configurations. Fast measurement times and easy handling make it viable for everyday QA and provide a means of independent source strength validation for pre-assembled plaques.

Materials and Methods

A Dexella 1521 CMOS flat-panel x-ray detector [Varex Imaging, Salt Lake City, USA] with a 75 um pixel size is used in conjunction with a 6 mm high, dome-shaped acrylic plaque holder and an in-house manufactured acrylic eye plaque, loaded with either Pd-103 seeds [IAPd-103, IsoAid, Port Richey, USA] or I-125 seeds [AgX100, Theragenics, Buford, CA]. For calibration of the detector and validation of the method, individual source strengths were measured with an ADCL calibrated HDR 1000 PLUS well chamber [Standard Imaging, Middleton, USA] before plaque assembly. *egs_brachy*, a brachytherapy specialized application for the EGSnrc radiation transport software, is used for MC simulation of the eye plaque dose distribution in the plane of the detector. The detector is calibrated by spatially registering the geometric central region of the measured and MC-calculated 2D dose distributions of three calibration eye plaques per source type, each loaded with various configurations of seeds.

Results

The calibration shows a linear dose response of the detector. Using this, the measured dose distributions of various plaque configurations show good agreement with MC calculations. The central dose rate is predicted by the measurements to within +-7% of the MC calculated central dose rate, allowing independent verification of the air-kerma strength of the assembled plaque. In addition, gamma-index analysis is used to compare measurements and MC calculations in the plane of the detector. This enabled the detection of individual mispositioned or dead seeds, as well as seeds that are out of specification enough to affect the dose distribution of the plaque.